

Tutorial V – Payload-Range Diagram

The economic potential and environmental footprint of an aircraft concept is crucial for success in the aviation market. Calculating mission performance, in the form of the payload-range diagram early in the design process can aid the selection of an aircraft configuration, tuning of the aircraft geometry and sizing of the propulsion system.



Basics

- 📖 How does a simplified mission for transport aircraft look like (including 6 segments)? Which segments are added according to Roskam's weight fraction definition? What else has to be considered in aircraft design?
- 📖 What is the difference between m_{OE} , $m_{TO,max}$ and $m_{ZF,max}$?
- 📖 What is depicted on a payload-range diagram? What do pilots, airlines and aircraft manufacturers use it for?
- 📖 Derive the Breguet equations for range and endurance for a turbojet in flight condition I (constant airspeed and lift coefficient).

Calculation tasks

- 1) Calculate the thrust-specific fuel consumption $TSFC$ of the engines during the cruise and the holding (loiter) flight segment. Use the closest meter value from the standard atmosphere sheet when converting a Flight Level (FL) to meters.
- 2) Examine the payload-range diagram on the next page. What payload mass in kg can be transported in each one of the points A, B and C? Define the take-off mass m_{TO} and landing mass m_L for each point. (Fuel reserves at landing can be neglected in a first approximation).
- 3) What is the tanked fuel fraction m_{fuel}/m_{TO} and the actual fuel mass m_{fuel} in each point? Use the relation $m_{fuel} = m_{TO} - m_L$ assuming all fuel is used prior to landing.
- 4) Calculate the weight fraction during the holding (loiter) segment w_{76} .
- 5) Calculate the weight fraction during the cruise segment w_{54} in each point by using the empirical mass fractions and the loiter mass fraction in combination with the individual calculated take-off and landing masses. Remember that: $m_{TO} \cdot (w_{10} \cdot w_{21} \cdot w_{32} \cdot w_{43} \cdot w_{54} \cdot w_{65} \cdot w_{76} \cdot w_{87}) = m_L$.
- 6) Finally, derive the cruise range from the weight fraction w_{54} for each point A, B, C and draw the payload-range diagram.

Assumptions and Additional Data

Cruise range conditions: $M = 0.78$ at $FL = 330$, $C_L/C_D = 16.3$

Holding (loiter) conditions: $M = 0.40$ at $FL = 100$, $C_L/C_D = 10$, $\Delta t = 45 \text{ min.}$

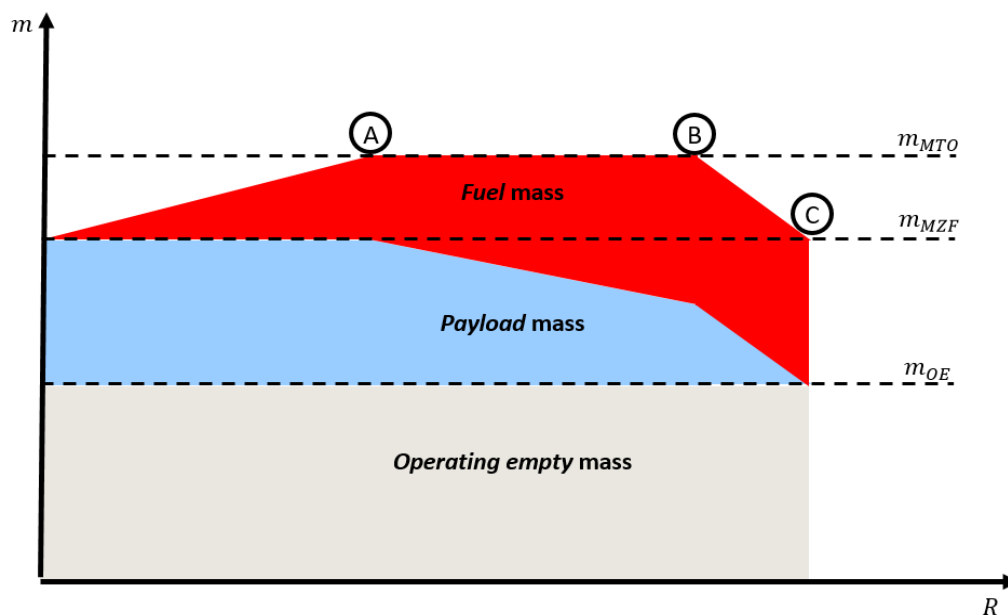
Engine specifications: $BPR = 9$, $c' = 0.020 \frac{\text{kg/s}}{\text{kN}}$

Mass properties:

- $m_{TO,max} = 73500 \text{ kg}$ (Maximum Take-Off mass)
- $m_{ZF,max} = 60500 \text{ kg}$ (Maximum Zero-Fuel mass) $= m_{OE} + m_{PL,max}$
- $m_{PL,max} = 19200 \text{ kg}$ (Maximum Payload mass)
- $m_{PL@fuel,max} = 13500 \text{ kg}$ (Maximum Payload mass at Max. Fuel tanked)
- $m_{OE} = 41300 \text{ kg}$ (Operating Empty mass)
- $m_{fuel,max} = 18700 \text{ kg}$ (Max. Fuel tank capacity)

Gravitational acceleration: $g = 9.81 \text{ m/s}^2$

Conversion FL to feet: $FL 1 = 100 \text{ ft}$



	Engine start & warm-up	Taxi	Take-off	Climb	Descent	Landing, taxi, shutdown
	W_{10}	W_{21}	W_{32}	W_{43}	W_{65}	W_{87}
Twin-engine, propeller	0.992	0.996	0.998	0.990	0.992	0.992
Regional turboprop	0.990	0.995	0.995	0.985	0.985	0.995
Transport jet	0.990	0.990	0.995	0.980	0.990	0.995
Business jets	0.990	0.995	0.995	0.980	0.990	0.992

For the cruise and holding (loiter) segments, apply the equation of Breguet for $n_V = 0$.

Equation

$$TSFC_x = c' \cdot (1 - 0.15 \cdot BPR^{0.65}) \cdot \left[1 + 0.28 \cdot (1 + 0.063 \cdot BPR^2) \cdot M_{\infty,x} \right] \cdot \left(\frac{\rho_x}{\rho_0} \right)^{0.08}$$

